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A nuclear reactor core (1) for a molten salt nuclear reactor (100), with a tubular cylindrical center moderator vessel (10) for passage of a liquid moderator (11) axially extending from a liquid moderator inlet (12) to a liquid moderator outlet (13), a plurality of tubular fuel salt vessels (20), each tubular fuel salt vessel (20) extending axially from a fuel salt inlet (22) to a fuel salt outlet (23) for passage of a molten fuel salt (21), the plurality of tubular fuel salt vessels (20) being assembled to form a cylindrical fuel salt jacket surrounding the center moderator vessel (10), a plurality of tubular neutron reflector vessels (30), each tubular neutron reflector vessel (30) extending axially from a liquid neutron reflector inlet (32) to a liquid neutron reflector outlet (33) for passage of a liquid neutron reflector (31), the plurality of tubular neutron reflector vessels (30) being assembled to form a cylindrical neutron reflector jacket surrounding the cylindrical fuel salt jacket.

Fortsættes...

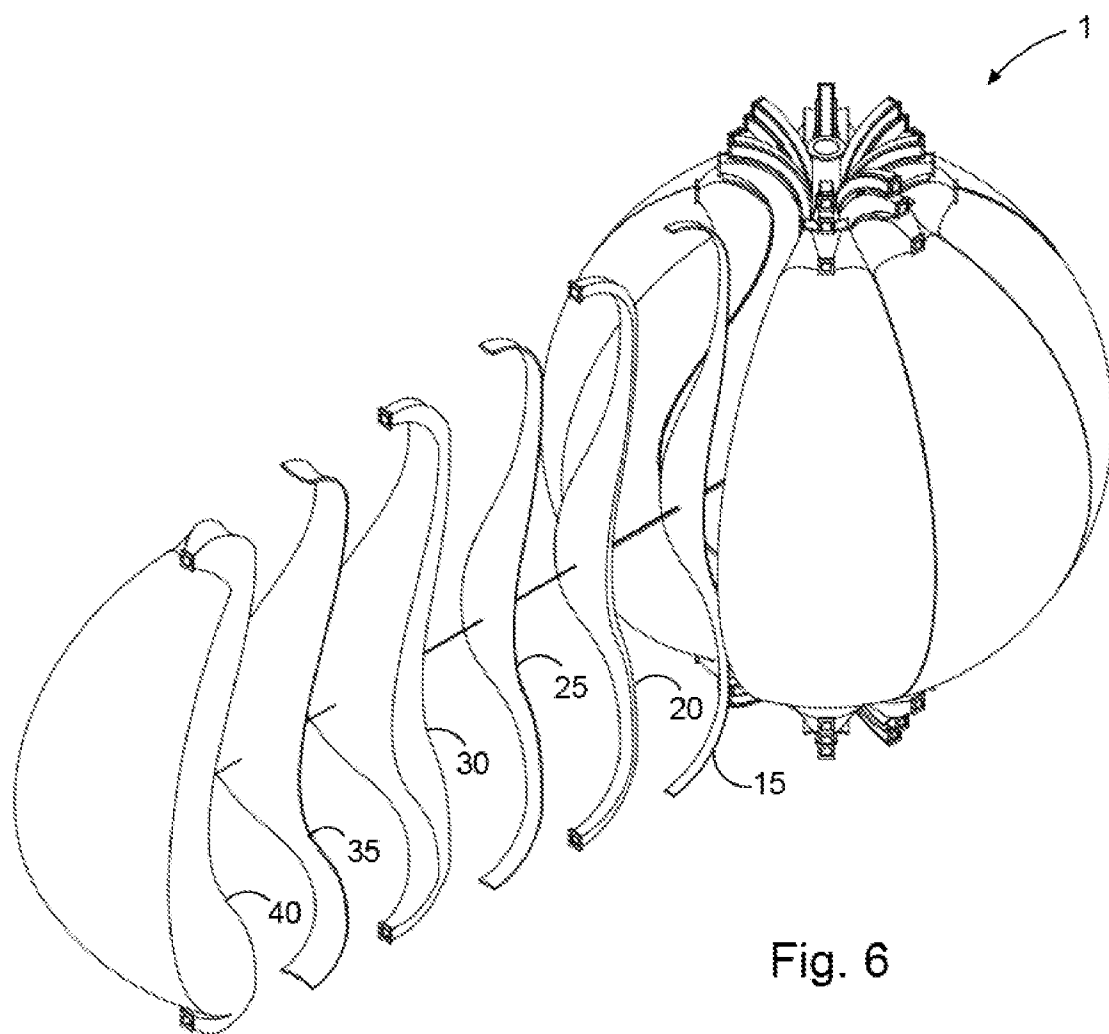


Fig. 6

TECHNICAL FIELD

The disclosure relates to a molten salt nuclear reactor, in particular to the construction and design of the nuclear reactor core of a molten salt nuclear reactor.

BACKGROUND

A molten salt reactor (MSR) is a nuclear reactor where the nuclear reactor coolant and/or the nuclear fuel is a molten salt, typically a fluoride or chloride salt, with a melting point of around ~ 500 °C, operating temperature of around ~ 600 to 700 °C, and a boiling point of ~ 1000 °C above the melting point. One of the many advantages of this type of reactor is that molten salts can be used as the heat transfer media at very high temperatures while still operating at or close to atmospheric pressure. Heat can be extracted from such reactors by pumping the molten salt in a loop between the nuclear reactor core and a heat exchanger with the reactor power being directly proportional to the temperature drop across the heat exchanger and the flow rate. Due to the corrosive nature of molten fluoride and chloride salt, their operation requires an inert containment atmosphere, furthermore molten salt or molten salt vapors cannot be allowed to escape to the environment, putting strict requirements on molten salt reactor components to be completely leak tight. This poses a severe technical challenge, since the temperature, the aggressive nature of the molten salt combined with high radiation levels renders only very few suitable materials to work with. Thus, the materials that can be used to construct the core have to be carefully selected and combined in order

to obtain a solution that provides a reliable and durable nuclear reactor core.

5 Molten salt reactors were built and operated at Oak Ridge National Laboratory (ORNL) in the 1950s and 1960s with a research program lasting to the 1970s and other small programs around the world.

10 The nuclear reactor core has a special geometry designed to allow a nuclear chain reaction to take place, achieved by either

1. a large enough amount of fuel to make a critical assembly or by
2. combining enough moderator and fuel to make a critical
15 assembly.

These are respectively called fast and thermal reactors because of the resulting neutron spectrum that each type exhibits.

20 There is a need for compact and mass manufacturable molten salt breeder reactors in order to achieve the goal of meeting the future global energy demand in a sustainable fashion. This creates a significant challenge since the smaller one makes a reactor the harder it becomes for it to achieve
25 breeding, since the probability of neutrons leaving the reactor, referred to as neutron leakage, roughly increases with the larger surface area to volume ratio of a small reactor.

30 Fast reactors generally require a much larger fissile inventory to become critical when compared to thermal reactors

and are thus not well suited to scale up rapidly to meet future energy demand because of limited availability of fissile material. Thus, a compact and mass manufacturable thermal molten salt breeder reactor is desired.

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One of the challenges for thermal molten salt breeder reactors is the need for a moderator that effectively slows down neutrons while allowing for breeding, which means that only moderators based on carbon, beryllium, or deuterium can be
10 used. The only practical moderators that will allow breeding in a thermal spectrum molten salt reactor are: solid carbon, solid beryllium, molten enriched lithium 7 deuterioxide salt (${}^7\text{LiOD}$), or liquid heavy water (D_2O). Of these, carbon is the only one that can withstand direct contact with the fuel salt,
15 while the others need to have a structural material separating them from the fuel salt. All these moderators have their pros and cons and have been proposed and studied in the past for use as moderators in a molten salt reactor.

20 Another challenge is the choice of materials for the vessels that contain the fuel salt or moderator material if the moderator is separated from the fuel salt. The materials need to resist degradation under extremely high temperatures, intense radiation exposure, and must have suitable neutron
25 absorption properties as well as resistance to the corrosive effect of molten salt for the vessels containing molten salt. Various materials have been proposed and studied in the past for use in the construction of the components of a molten salt reactor. However, each of these materials has practical
30 limitations in relation to the shapes that are possible to produce in commercial manufacturing.

The only known molten salt compatible materials with low neutron absorption are carbon and silicon carbide based materials. The most promising of these are the ceramic
5 composites of silicon carbide and carbon, e.g. carbon fiber-carbon matrix C/C composite, silicon carbide fiber-silicon carbide matrix (SiC/SiC) composite, and silicon carbide fiber-carbon matrix (SiC/C) composite. The construction of complex shaped composite materials is much more challenging
10 than from metals since these complex shaped composite materials cannot be bent and or welded in their final form but rather have to be molded to shape and treated in multiple steps, e.g. furnace firing.

15 CN112259263A discloses a nuclear reactor core according to the preamble of claim 1 with a construction with straight cylindrical vessels, that is relatively easy to construct from suitable materials, but does not provide an effective reactor design that minimizes neutron leakage.

20 SUMMARY

It is an object to provide a nuclear reactor core for a nuclear reactor that overcomes or at least reduces the problem above.

25 The foregoing and other objects are achieved by the features of the independent claims. Further implementation forms are apparent from the dependent claims, the description and the figures.

According to a first aspect, there is provided nuclear reactor core for a molten salt nuclear reactor, the nuclear reactor core comprising:

- a tubular cylindrical center moderator and neutron reflector vessel for passage of a liquid moderator and neutron reflector axially extending from a liquid moderator and neutron reflector inlet to a liquid moderator and neutron reflector outlet,

- a plurality of tubular fuel salt vessels, each tubular fuel salt vessel extending axially from a fuel salt inlet to a fuel salt outlet for passage of a molten fuel salt, the plurality of tubular fuel salt vessels being assembled to form a cylindrical fuel salt jacket surrounding the center moderator and neutron reflector vessel,

- a plurality of tubular moderator and neutron reflector vessels, each tubular moderator neutron reflector vessel extending axially from a liquid moderator and neutron reflector inlet to a liquid moderator and neutron reflector outlet for passage of a liquid moderator and neutron reflector, the plurality of tubular moderator and neutron reflector vessels being assembled to form a cylindrical moderator and neutron reflector jacket surrounding the cylindrical fuel salt jacket, and

- the cylindrical center moderator and neutron reflector vessel has a largest inner cross-sectional area medially between the liquid moderator and neutron reflector inlet of the center moderator and neutron reflector vessel and the liquid moderator and neutron reflector outlet of the moderator and neutron reflector vessel.

By constructing the nuclear reactor core with a fuel salt jacket that is formed by a plurality of tubular vessels that, when assembled, form the fuel salt jacket and by constructing the nuclear reactor core with a neutron reflector jacket that is formed by a plurality of tubular vessels than when assembled form the neutron reflector jacket, it becomes possible to construct a nuclear reactor core with a design of concentric cylinders that can have a varying radius to allow a larger volume in the respective concentric spaces between the respective cylinders in which the moderator and reflector liquid and the molten fuel salt is received for achieving a critical assembly and reducing neutron leakage, i.e. an efficient reactor geometry, while still being able to manufacture from materials that are suitable for use in a molten salt nuclear reactor. The components are made from materials that typically need to be made in a mold which in turn poses a range of restrictions in the design for these components. The core construction according to the present invention allows these components to be designed in such a way that they can be manufactured from suitable materials, such as ceramics composite and metal alloy, preferably zirconium alloy. The construction of the nuclear reactor core according to the first aspect allows for an optimized core design with an efficient reactor geometry where only a small amount of fuel salt is needed to create a critical reactor assembly while still being practical to assemble and also allowing for a sufficient fuel salt circulation rate to extract the produced heat, with the cylindrical center moderator and neutron reflector vessel having a largest inner cross-sectional area medially between the liquid moderator inlet and the liquid moderator and neutron reflector outlet.

In a possible implementation of the first aspect, the ceramic composite material is a ceramic matrix composite.

5 In a possible implementation of the first aspect, the ceramic composite material is a ceramic composite of silicon carbide and carbon, preferably a carbon fiber-carbon matrix (C/C) composite, silicon carbide fiber-silicon carbide matrix (SiC/SiC) composite, and/or silicon carbide fiber-carbon
10 matrix (SiC/C) composite.

In a possible implementation form of the first aspect, the inner cross-sectional area of the center moderator and neutron reflector vessel increases gradually towards the medial
15 position between the liquid moderator and neutron reflector inlet and the liquid moderator and neutron reflector outlet. The inner cross-sectional area preferably varying continuously between the moderator and neutron reflector inlet and the moderator and neutron reflector outlet.

20

In a possible implementation form of the first aspect, the cylindrical center moderator and neutron reflector vessel and/or the liquid moderator and neutron reflector vessels have a wall of a metal alloy, preferably iron alloy,
25 preferably stainless steel and most preferably zirconium alloy or ceramic composite material, and wherein the fuel salt vessels preferably having a wall of ceramic composite material, a metal alloy, an iron alloy, preferably stainless steel, or nickel alloy, preferably Inconel® or Hastelloy®.

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In a possible implementation form of the first aspect, each fuel salt vessel in the fuel salt jacket is substantially identical to the other fuel salt vessels in the fuel salt jacket.

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In a possible implementation form of the first aspect, each moderator and neutron reflector vessel in the moderator and neutron reflector jacket is substantially identical to the other moderator and neutron reflector vessels in the moderator and neutron reflector jacket.

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In a possible implementation form of the first aspect, a first cylindrical insulation jacket is provided between the center moderator vessel and the fuel salt jacket, the first insulation jacket preferably being formed by one or more first insulation members, the first insulation members preferably being shaped like sectors of the first cylindrical insulation jacket or shaped as a strip that is spirally wound around the center moderator vessel, and the first insulation members preferably made being of a graphite felt or insulation aerogel impregnated graphite felt. The insulation allows a much lower heat loss to the moderator and neutron reflector liquid and for the moderator and neutron reflector liquid to be consistently kept operating at a much lower temperature than the fuel salt, which is in particularly a significant advantage if a hydroxide or deuterioxide moderator is used, due to the reduced corrosive effect at lower temperatures.

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In a possible implementation form of the first aspect, the second cylindrical insulation jacket is provided between the fuel salt jacket and the moderator and neutron reflector

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jacket, the second insulation jacket preferably being formed by one or more second insulation members, the second insulation members preferably being shaped like sectors of the second cylindrical insulation jacket or shaped as a strip
5 that is spirally wound around the fuel salt jacket, and the second insulation members preferably being made of a graphite felt or insulation aerogel impregnated graphite felt.

In a possible implementation form of the first aspect, the
10 nuclear reactor core comprises a plurality of tubular blanket salt vessels, each tubular blanket salt vessel extending axially from an inlet to an outlet for passage of a molten blanket salt, the plurality of tubular blanket salt vessels being assembled to form a cylindrical blanket salt jacket
15 surrounding the moderator and neutron reflector jacket.

In a possible implementation form of the first aspect, a third cylindrical insulation jacket is provided between the neutron reflector vessel and the blanket salt jacket, the third
20 insulation jacket preferably being formed by one or more third insulation members, the third insulation members preferably being shaped like sectors of the third cylindrical insulation jacket or shaped as a strip that is spirally wound around the moderator and neutron reflector jacket, and the third
25 insulation members preferably being made of a graphite felt or insulation aerogel impregnated graphite felt.

In a possible implementation form of the first aspect, the center moderator and neutron reflector vessel, the fuel salt
30 vessels, the moderator and neutron reflector vessels, and/or the blanket salt vessels are each assembled from two parts

that are separated by a symmetry plane M that extends radially from the medial position.

In a possible implementation form of the first aspect, the center moderator and neutron reflector vessel has a variable inner and outer radii R11 and R12, the inner and outer Radii R11 and R12 preferably being largest at or near the medial position and gradually decreasing from the medial position.

In a possible implementation form of the first aspect, the fuel salt jacket is shaped as a hollow circular cylinder with variable inner and outer radii R21 and R22, and wherein the fuel salt vessels are preferably shaped as a circular cylindrical sector with variable inner and outer radii R21 and R22, respectively, and a given sector angle A2, the inner and outer Radii R21 and R22 preferably being largest at or near the medial position and gradually decreasing from the medial position.

In a possible implementation form of the first aspect, the moderator and neutron reflector jacket is shaped like a hollow circular cylinder with variable inner and outer radii R31 and R32, and wherein the moderator neutron reflector vessel are preferably shaped like a circular cylindrical sector with variable inner and outer radii R31 and R32, respectively, and a given sector angle A3, the inner and outer Radii R31 and R32 preferably being largest at or near the medial position and gradually decreasing from the medial position.

In a possible implementation form of the first aspect, the blanket salt jacket is shaped as a hollow circular cylinder

with variable inner and outer radii R_{41} and R_{42} , and wherein the blanket salt vessels is shaped like a circular cylindrical sector with variable inner and outer radii R_{41} and R_{42} , respectively, and a given sector angle A_4 , the inner and outer
5 Radii R_{41} and R_{42} preferably being largest at or near the medial position and gradually decreasing from the medial position.

In a possible implementation form of the first aspect, the
10 fuel salt jacket, the neutron reflector jacket, and/or the blanket jacket are circumferentially potentially continuous jackets, preferably formed by the matching sides of the tubular vessels forming these jackets touching one another along a substantial portion of their axial extent.

15 In a possible implementation form of the first aspect, the center moderator vessel is segmented and formed by a plurality of tubular moderator vessels, each moderator vessel extending axially from a moderator inlet to a moderator outlet for
20 passage of a moderator, the plurality of tubular moderator vessels being assembled to form the center moderator vessel.

In a possible implementation form of the first aspect, the center moderator vessel has a circumferential outer wall and
25 a circumferential inner wall, the circumferential inner wall preferably creating a lumen for receiving a control rod.

In a possible implementation form of the first aspect, the liquid moderator is heavy water.

In a possible implementation form of the first aspect, the fuel salt comprises fissile components, preferably comprising enriched lithium 7 fluoride, thorium tetrafluoride, uranium tetrafluoride, uranium trifluoride and/or plutonium trifluoride (7LiF)-(ThF₄)-(UF₄)-(UF₃)-(PuF₃) salt.

In a possible implementation form of the first aspect, the blanket salt is a molten salt comprising fertile components, preferably comprising enriched lithium 7 fluoride and/or thorium tetrafluoride (7LiF)-(ThF₄) salt.

In a possible implementation form of the first aspect, the nuclear reactor core is provided with movable neutron absorbing control elements between vessels of the jackets, preferably between vessels making up the fuel salt jacket and/or between vessels forming a jacket that surrounds the fuel jacket or is surrounded by the fuel jacket. The neutron absorbing control elements being similar to control rods, except that the neutron absorbing control elements are not rod shaped but are shaped like a plate or wedge to match the adjacent vessels in the jacket concerned, and are arranged such that they can be inserted into and retracted from the nuclear reactor core, for controlling nuclear reactivity of the nuclear reactor core.

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In a possible implementation form of the first aspect a second fuel salt jacket surrounds the moderator and neutron reflector jacket, and a second moderator and neutron reflector jacket surrounds the second fuel salt jacket. In this implementation, the blanket jacket, if present, surrounds the second neutron reflector jacket. In this implementation, the second fuel

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salt jacket is formed by a plurality of tubular fuel salt vessels, each tubular fuel salt vessel extending axially from a fuel salt inlet to a fuel salt outlet for passage of a molten fuel salt, the plurality of tubular fuel salt vessels
5 being assembled to form a second cylindrical fuel salt jacket surrounding the moderator and neutron reflector jacket. The second neutron reflector jacket is formed by a plurality of tubular moderator and neutron reflector vessels. Each tubular moderator neutron reflector vessel extending axially from a
10 liquid moderator and neutron reflector inlet to a liquid moderator and neutron reflector outlet for passage of a liquid moderator and neutron reflector.

According to a second aspect, there is provided a method of
15 operating a nuclear reactor core of a molten salt nuclear reactor, the nuclear reactor core comprising:

- a tubular cylindrical center moderator and neutron reflector vessel for passage of a liquid moderator and neutron reflector axially extending from a liquid moderator and neutron
20 reflector inlet to a liquid moderator and neutron reflector outlet,

- a cylindrical fuel salt jacket surrounding the center moderator vessel, the fuel salt jacket extending axially from a fuel salt inlet to a fuel salt outlet for passage of a
25 molten fuel salt, and

- a cylindrical moderator and neutron reflector jacket surrounding the fuel salt jacket, the moderator and neutron reflector jacket extending axially from liquid moderator and neutron reflector inlet to a moderator and neutron reflector
30 outlet for passage of a liquid moderator and neutron reflector,

the method comprising:

-controlling temperature of the fuel salt in the fuel salt jacket to remain at a temperature between approximately 600 and 700°C,

5 -controlling the temperature of the moderator and neutron reflector liquid in the central moderator and neutron reflector vessel to remain at a temperature between approximately 20 and 90°C, and

-preferably controlling the temperature of the moderator and
10 neutron reflector liquid in the moderator and neutron reflector jacket to remain at a temperature between approximately 20 and 90°C.

By controlling the temperature of the moderator between the
15 approximately 20 and the person 90°C, the moderator and neutron reflector, typically heavy water, does not need to be pressurized to prevent it from boiling. Thus, the vessels that contain the moderator and neutron reflector will not need to act as a pressure vessel and can thus be constructed
20 in a much lighter and less solid way. This is a significant advantage, since the suitable materials that have the required properties are difficult for use in a construction that needs to be withstand pressure (pressure vessel). Thus, the construction of the nuclear core is significantly
25 facilitated.

In a possible implementation form of the second aspect, the method comprises:

-adjusting the liquid level of the liquid moderator and

30 neutron reflector in the center moderator and neutron reflector vessel, and/or

- adjusting the liquid level of the liquid moderator and neutron reflector in the moderator and neutron reflector jacket, and/or
- adjusting the liquid level of the molten blanket salt in the
5 blanket salt jacket, and/or
- adjusting temperature of fuel salt in the fuel jacket, and/or
- adjusting temperature of the liquid moderator and neutron reflector in the center moderator and neutron reflector vessel, and/or
- 10 -adjusting temperature of the liquid moderator and neutron reflector in the moderator and neutron reflector jacket, and/or
- adjusting the chemical composition of the liquid moderator and neutron reflector, and/or
- 15 -adjusting the chemical composition of the fuel salt, and/or
- adjusting flow rate of the liquid moderator and neutron reflector in the center moderator and neutron reflector vessel, and/or
- adjusting the position of a control rod that is at least
20 partially inserted in a lumen in the center moderator and neutron reflector vessel, and/or
- adjusting the position of a control rod that is at least partially inserted between fuel salt vessels, and/or
- adjusting the position of a control rod that is at least
25 partially inserted between moderator and neutron absorption vessels.

In a possible implementation form of the second aspect, the control rod is of a neutron absorbing material such as boron
30 or hafnium.

In a possible implementation form of the second aspect, the liquid moderator is heavy water.

5 In a possible implementation form of the second aspect, the fuel salt comprises fissile components, preferably comprising enriched lithium 7 fluoride, thorium tetrafluoride, uranium tetrafluoride, uranium trifluoride and/or plutonium trifluoride (7LiF)-(ThF₄)-(UF₄)-(UF₃)-(PuF₃) salt.

10 In a possible implementation form of the second aspect, the blanket salt is a molten salt comprising fertile components, preferably comprising enriched lithium 7 fluoride and/or thorium tetrafluoride (7LiF)-(ThF₄) salt.

15 According to a third aspect there is provided a molten salt nuclear reactor comprising a controller configured to perform the method according to the second aspect or any possible implementations of the second aspect.

20 According to a fourth aspect, there is provided nuclear reactor core of a molten salt nuclear reactor, the nuclear reactor core comprising:

- a tubular cylindrical center moderator and neutron reflector vessel for passage of a liquid moderator and neutron reflector
- 25 axially extending from a liquid moderator and neutron reflector inlet to a liquid moderator and neutron reflector outlet, the center moderator and neutron reflector vessel having a wall of metal alloy, preferably zirconium alloy or ceramic composite material,
- 30 - a cylindrical fuel salt jacket surrounding the center moderator vessel, the fuel salt jacket extending axially from

a fuel salt inlet to a fuel salt outlet for passage of a molten fuel salt, the fuel salt jacket having walls of ceramic composite material, and

a cylindrical moderator and neutron reflector jacket
5 surrounding the fuel salt jacket, the moderator and neutron reflector jacket extending axially from liquid moderator and neutron reflector inlet to a moderator and neutron reflector outlet for passage of a liquid moderator and neutron reflector, the moderator and neutron reflector jacket having
10 walls of metal alloy, preferably zirconium alloy or ceramic composite material.

The choice of materials for the vessels, and the fourth aspect ensures relatively low neutron absorption by the vessels while
15 still providing good tolerance towards the molten salts and moderator liquids.

In a possible implementation form of the fourth aspect, the nuclear reactor core comprises a first cylindrical insulation
20 jacket between the center moderator vessel and the fuel salt jacket, the first insulation jacket comprising graphite felt or insulation aerogel impregnated graphite felt, or other high temperature insulation material, such as ceramic fiber insulation.

25 In a possible implementation form of the fourth aspect, the nuclear reactor core comprises a second cylindrical insulation jacket between the fuel salt jacket and the moderator neutron reflector jacket, the second insulation
30 jacket comprising graphite felt or insulation aerogel impregnated graphite felt.

In a possible implementation form of the fourth aspect, the liquid moderator and neutron reflector is heavy water or a molten hydroxide, preferably molten enriched lithium 7 deuterioxide salt (${}^7\text{LiOD}$).

In a possible implementation form of the fourth aspect, the control rod comprises a neutron absorbing material, preferably boron or hafnium.

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In a possible implementation form of the fourth aspect, the fuel salt comprises fissile components, preferably comprising enriched lithium 7 fluoride, thorium tetrafluoride, uranium tetrafluoride, uranium trifluoride and/or plutonium trifluoride ${}^7\text{LiF-ThF}_4\text{-UF}_4\text{-UF}_3\text{-PuF}_3$ salt.

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In a possible implementation form of the fourth aspect, the blanket is a molten salt comprising fertile components, preferably comprising enriched lithium 7 fluoride and/or thorium tetrafluoride (${}^7\text{LiF-ThF}_4$) salt.

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According to a fifth aspect there is provided a molten salt nuclear reactor comprising a nuclear reactor core according to any the fourth aspect or any possible implementations of the fourth aspect.

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These and other aspects will be apparent from and the embodiments described below.

30 BRIEF DESCRIPTION OF THE DRAWINGS

In the following detailed portion of the present disclosure, the aspects, embodiments, and implementations will be explained in more detail with reference to the example embodiments shown in the drawings, in which:

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Fig. 1 is a diagrammatic representation of a molten salt reactor with a nuclear reactor core according to an embodiment;

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Fig. 2 is an elevated view of a nuclear reactor core according to an embodiment,

Fig. 3 is a sectional view of the nuclear reactor core of Fig. 2 including an enlarged detail,

Fig. 4 is a cross-sectional view of the nuclear reactor core of Fig. 2 including an enlarged detail,

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Fig. 5 is an elevated and exploded view of a portion of the nuclear reactor core of Fig. 2, and

Fig. 6 is an elevated and exploded view of a portion of the nuclear reactor core of Fig. 2 together with the remaining non-exploded portions of the nuclear reactor core.

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DETAILED DESCRIPTION

Fig. 1 illustrates an embodiment of a molten salt nuclear reactor 100 using an embodiment of the nuclear reactor core 1. The molten salt nuclear reactor 100 uses a plurality of flow machines 1 (pumps) for circulating the molten fuel salt. Each flow machine 21 is connected to a heat exchanger 3 via a fuel salt loop 4, a reactor coolant loop 5 and a secondary coolant loop 6 for heat exchanging with the nuclear reactor core 1. The fuel salt loop 4 provides fuel salt for driving and controlling the nuclear reaction. Further, the heat exchangers 3 provide the reactor coolant and secondary

coolants via the fuel salt loop 4 and the reactor coolant loop 5 and secondary coolant loop 6. In order to circulate and drive the flow of molten salt, flow machines 1 (pumps) are used.

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With reference to Figs. 2 to 6, an embodiment of the nuclear reactor core 1 is shown with an inlet and/or outlet area 7 at the upper end and an inlet and/or outlet area 8 at the lower end.

10

The nuclear reactor core 1 has an axis X and extends along the axis X extends between the inlet area 7 and the outlet area 8.

15 In the shown embodiment, the nuclear reactor core 1 is cylindrical and almost spherical. The nuclear reactor core 1 has a cylindrical and concentric construction. A tubular cylindrical center moderator and neutron reflector vessel 10 extends along the axis X. The cylindrical center moderator and neutron reflector vessel 10 provides for passage of a liquid moderator and neutron reflector 11 between a liquid moderator and neutron reflector inlet 12 and a liquid moderator and neutron reflector outlet 13. The liquid moderator and neutron reflector can be heavy water or a molten hydroxide, such as molten enriched lithium 7 deuterioxide salt (7LiOD).

The cylindrical center moderator and neutron reflector vessel 10 has a largest inner cross-sectional area medially between the liquid moderator inlet 12 and the liquid moderator and neutron reflector outlet 13. The inner cross-sectional area

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of the center moderator and neutron reflector vessel 10 preferably increasing gradually towards a medial position between the liquid moderator and neutron reflector inlet 12 and the liquid moderator and neutron reflector outlet 13 the inner cross-sectional area of the center moderator and neutron reflector vessel 10 preferably varying continuously between the moderator and neutron reflector inlet 12 and the moderator and neutron reflector outlet 13. Thus, the center moderator and neutron reflector vessel 10 has a varying inner radius. The inner radius is varied in a way such as to achieve a semi-spherical shape of the cylindrical center moderator and neutron reflector vessel 10. The center moderator and neutron reflector vessel 10 has variable inner and outer radii R11 and R12, the inner and outer Radii R11 and R12 preferably being largest at or near the medial position and gradually decreasing from the medial position. During operation of the nuclear reactor core 1, the center moderator and neutron reflector vessel 10 is at least partially filled or completely with a liquid moderator neutron reflector 11, and the liquid moderator and neutron reflector 11 is exchanged at a controlled rate by flow through the moderator and neutron reflector vessel 10.

The cylindrical center moderator and neutron reflector vessel 10 is in an embodiment a wall of a metal alloy, preferably iron alloy (when the moderator and neutron absorption liquid is heavy water), preferably stainless steel, and most preferably zirconium alloy or ceramic composite material. The ceramic composite material is in an embodiment ceramic matrix composite. The ceramic matrix composite comprises ceramic fibers embedded in a ceramic matrix. The fibers and the matrix

both can consist of any ceramic material, whereby carbon and carbon fibers can also be regarded as a ceramic material. The ceramic composite material is in an embodiment a composite of silicon carbide and carbon, e.g. carbon fiber-carbon matrix (C/C) composite, silicon carbide fiber-silicon carbide matrix (SiC/SiC) composite, and or silicon carbide fiber-carbon matrix (SiC/C) composite.

A cylindrical fuel salt jacket extends along the axis X between the inlet and outlet area 7,8 and surrounds the center moderator and neutron reflector vessel 10. The fuel salt jacket is formed by assembling a plurality of tubular fuel salt vessels 20. Each tubular fuel salt vessel 20 extends axially from a fuel salt inlet 22 to a fuel salt outlet 23 for passage of a molten fuel salt 21. Each fuel salt vessel 20 in the fuel salt jacket is substantially identical to the other fuel salt vessels 20 in the fuel salt jacket. During operation of the nuclear reactor core 1, the fuel salt jacket 20 is at least partially or completely filled with fuel salt 21, and the fuel salt 21 is exchanged at a controlled rate by flow through the fuel salt vessels 20.

A first cylindrical insulation jacket is provided between the center moderator vessel 10 and the fuel salt jacket. The first insulation jacket is formed by one or more first insulation members 15. The first insulation members 15 can, as shown, be shaped like sectors of the first cylindrical insulation jacket. Alternatively, the first insulation members 15 can be shaped as a strip that is spirally wound around the center moderator vessel 10. The first insulation members 15 are in an embodiment made of a graphite felt or insulation aerogel

impregnated graphite felt. The first insulation jacket 15 allows a much lower heat loss to the moderator and neutron reflector liquid 11 and for the moderator and neutron reflector liquid 11 to be consistently kept operating at a much lower temperature than the fuel salt.

The cylindrical fuel salt jacket is shaped to match the shape of the cylindrical center moderator and neutron reflector vessel 10 (with the first insulation jacket therebetween), i.e. complementary therewith. Hence, the inner radius of the fuel salt jacket varies in a similar way to the outer radius of the cylindrical center moderator and neutron reflector vessel 10, such that the first insulation jacket 15 fits in between.

The fuel salt jacket is shaped as a hollow circular cylinder with variable inner and outer radii R_{21} and R_{22} . The fuel salt vessels 20 are preferably shaped as a circular cylindrical sector with variable inner and outer radii R_{21} and R_{22} , respectively, and a given sector angle A_2 , the inner and outer Radii R_{21} and R_{22} preferably being largest at or near the medial position and gradually decreasing from the medial position. In the present embodiment, the sector angle is the same for all elements of the nuclear reactor core 1 and hence the sector angle has been indicated in Fig. 4 as " A_n ". However, it should be understood that the sector angle " A_n " does not need to be the same for the various jackets and not even for the various vessels that make up a single jacket.

The fuel salt jacket is a circumferentially substantially continuous jacket that is formed by the matching sides of the

tubular fuel salt vessels 20 touching one another along a substantial portion of their axial extent.

A cylindrical moderator and neutron reflector jacket
5 surrounds the cylindrical fuel salt jacket. The cylindrical moderator and neutron reflector jacket is formed by assembling a plurality of tubular moderator and neutron reflector vessels 30. Each tubular moderator neutron reflector vessel 30 extends axially from a liquid moderator and neutron reflector inlet
10 32 to a liquid moderator and neutron reflector outlet 33 for passage of a liquid moderator and neutron reflector 31. The elements that make up the tubular moderator neutron reflector vessel 30 have walls of ceramic composite material or metal alloy, preferably zirconium alloy. The ceramic composite
15 material is in an embodiment ceramic matrix composite. The ceramic matrix composite comprises ceramic fibers embedded in a ceramic matrix. The fibers and the matrix both can consist of any ceramic material, whereby carbon and carbon fibers can also be regarded as a ceramic material. The ceramic composite
20 material is in an embodiment a composite of silicon carbide and carbon, e.g. carbon fiber-carbon matrix (C/C) composite, silicon carbide fiber-silicon carbide matrix (SiC/SiC) composite, and or silicon carbide fiber-carbon matrix (SiC/C) composite. In an embodiment each moderator neutron reflector
25 vessel 30 in the moderator neutron reflector jacket 30 substantially identical to the other moderator neutron reflector vessels 30 in the fuel salt jacket.

The moderator and neutron reflector jacket is shaped as a
30 hollow circular cylinder with variable inner and outer radii R31 and R32. The moderator neutron reflector vessels 30 are

shaped as a circular cylindrical sector with variable inner and outer radii R_{31} and R_{32} , respectively, and a given sector angle A_3 . The inner and outer Radii R_{31} and R_{32} are largest at or near the medial position and gradually decrease from
5 the medial position.

During operation of the nuclear reactor core 1, the moderator and neutron reflector jacket is at least partially filled or completely with a liquid moderator neutron reflector 31, and
10 the liquid moderator and neutron reflector 31 is exchanged at a controlled rate by flow through the moderator and neutron reflector vessels 30.

A second cylindrical insulation jacket is provided between
15 the fuel salt jacket and the moderator and neutron absorption jacket. The second insulation jacket is formed by one or more second insulation members 25. The second insulation members 25 can, as shown, be shaped like sectors of the second cylindrical insulation jacket. Alternatively, the second
20 insulation members 25 can be shaped as a strip that is spirally wound around the fuel salt jacket. The second insulation members 25 are in an embodiment made of a graphite felt or insulation aerogel impregnated graphite felt. The second insulation jacket allows a much lower heat loss to the
25 moderator and neutron reflector liquid 31 and for the moderator and neutron reflector liquid 31 to be consistently kept operating at a much lower temperature than the fuel salt 21.

30 An optional blanket salt jacket surrounds the moderator and neutron absorption jacket, preferably with a third insulation

jacket there between. The blanket salt jacket comprises a plurality of tubular blanket salt vessels 40, each tubular blanket salt vessel 40 extending axially from an inlet to an outlet for passage or be constructed without blanket salt inlet and outlet holding of a molten blanket salt 41 in a stationary fashion. The plurality of tubular blanket salt vessels 40 are assembled to form a cylindrical blanket salt jacket surrounding the moderator and neutron reflector jacket.

The blanket salt jacket is shaped as a hollow circular cylinder with variable inner and outer radii R_{41} and R_{42} . The blanket salt vessels 40 are shaped as a circular cylindrical sector with variable inner and outer radii R_{41} and R_{42} , respectively, and a given sector angle A_4 . The inner and outer Radii R_{41} and R_{42} are largest at or near the medial position and gradually decrease from the medial position. The blanket salt jacket can be constructed without blanket salt inlets and outlets, in which the blanket salt 41 is not continuously exchanged and could be either maintained molten, near its melting point to prevent radiolysis of the blanket salt, or frozen.

A third cylindrical insulation jacket is provided between the neutron reflector vessel and the blanket salt jacket. The third insulation jacket is preferably formed by one or more third insulation members 35. The third insulation members 35 can be shaped like sectors of the third cylindrical insulation jacket or shaped as a strip that is spirally wound around the moderator and neutron reflector jacket. The third insulation

members 35 are preferably made of a graphite felt or insulation aerogel impregnated graphite felt.

The fuel salt vessels 20, the moderator and neutron reflector vessels 30, and/or the blanket salt vessels 40 are, in an embodiment, each assembled from two parts that are separated by a symmetry plane M that extends radially from the medial position (radially relative to axis A). The inlet and outlet sections of these vessels 20, 30, and 40 can, as shown, curve radially outward and be provided with a flange or the like for connection to piping. However, it should be understood that the inlet and outlet sections can also extend axially and do not need to be provided with a flange for connection to piping since other solutions for connecting the vessels 20, 30, and 40 to piping are known in the art. The inlet and outlet 12,13 of the moderator and neutron reflector vessel 10 are in an embodiment provided with a flange for connection to pumping, but it is understood that the inlet and outlet 12,13 can be formed without a flange and be connected to piping by another form of assembly.

The vessels 20, 30, and 40 and possibly their inlet and outlet sections are held together by a structure that is not shown in the figures for reasons of simplicity. Such support structures are thought well known in the art.

The design of the nuclear reactor core 1 is an attempt to obtain a nuclear reactor core that is as spherical as achievable, but still constructible from materials that can withstand the harsh conditions in the nuclear reactor core 1 while allowing for a sufficient fuel salt circulation rate to

extract the produced heat. The result is a design with a structure that is both cylindrical and spherical, as well as both layered and segmented, i.e. resembling a hybrid of the layered structure and shape of onions and the segmented
5 structure and shape of citrus fruit.

Traditional ceramics are relatively brittle, have relatively low thermal shock resistance, and relatively low fracture toughness. Ceramic composite materials or ceramic matrix
10 composite materials are made from short, continuous, or braided ceramic fiber material, usually embedded in a ceramic matrix, providing reinforcement of the matrix ceramic. Ceramic composites subdue or at least reduce the drawbacks of traditional ceramics.

15 The fibers and the matrix are made from a variety of ceramic materials and in a variety of different processes resulting in differences in mechanical properties, such as strength and porosity, and impurity content. Most common commercially
20 available ceramic composite materials are made from carbon and/or silicon carbide fibers and carbon or silicon carbide matrix, abbreviated (C/C), (C/SiC), (SiC/C), and (SiC/SiC). Both naturally occurring carbon, mainly containing the carbon 12 isotope, and silicon, mainly containing silicon 28 isotope,
25 have low neutron capture probability in a thermal neutron energy spectrum. Furthermore, both carbon and silicon carbide have good corrosion resistance to the molten salt used in the molten salt reactor, i.e. both are corrosive resistant to molten salt and have low neutron absorption in the thermal
30 spectrum. However, both are too brittle to be used as a construction material for vessels and jackets in a molten

salt nuclear reactor core. Yet, their ceramic composites are interesting candidates for use as construction material in molten salt reactors because of their relatively high fracture toughness and high thermal shock resistance combined with
5 their low neutron absorption in thermal spectrum and exceptional resistance to radiation damage.

The construction of the vessels (moderator and neutron reflector vessel 10, and/or moderator and neutron reflector
10 vessels 30 and/or fuel salt vessels 20 and/or blanket salt vessels 40) from ceramic composites comprises lay-up and fixation, where ceramic fiber or resin impregnated ceramic fibers are wound or placed in or around a mold/core, giving the part the shape of the final ceramic composites part, then
15 polymer is infiltrated into the fibers, then the part is cured. After curing the part is demolded (removed from mold/core) and the polymer is pyrolyzed in a furnace at high temperatures, usually above 800°C in an inert atmosphere to form the matrix composite (in an embodiment the mold/core is
20 a one use mold/core that is burned, melted, crushed or destroyed after first use. This leaves the part porous and the part is re-infiltrated (resin impregnated) and the pyrolyzing step is repeated until the desired porosity is reached.

25

Several minor variations of the above process can be used, however, all these processes depend on the first step of shaping the fibers and polymer in a mold and/or on a core. This limits the geometry of the jackets, vessels and other
30 parts that can be manufactured, since the jackets, vessels and other parts of the nuclear reactor core 1 have to be able

to release from the mold or fixture (core) with the vessel in its final form, allowing for some intermediate and or final machining to tolerances. The process does not allow for creation of complex shapes compared to traditional metal manufacturing, such as casting, forming (e.g. forging, rolling, extruding, die forming, indenting, stretching, deep drawing, stamping and bending). Further, and machining (e.g. turning, drilling, boring, milling and cutting) but more technically challenging compared with machining conventional metal parts. However, the disclosed construction and geometry of the present nuclear reactor core allows each of the vessels and jackets to be manufactured by the commercially available processes for manufacturing parts in composite ceramic material.

In an embodiment (not shown in the Figs.), the center moderator vessel 10 is segmented and formed by a plurality of tubular moderator vessels, each moderator vessel extending axially from a moderator inlet to a moderator outlet for passage of a moderator, the plurality of tubular moderator vessels being assembled to form the center moderator vessel 10. In a variation of this embodiment, the center moderator vessel 10 has a circumferential outer wall and a circumferential inner wall, the circumferential inner wall creating a lumen for receiving a control rod. The control rod is of a neutron absorbing material such as boron or hafnium.

In an embodiment (not shown in the Figs.) a second fuel salt jacket surrounds the moderator and neutron reflector jacket, and a second moderator and neutron reflector jacket surrounds the second fuel salt jacket. In this embodiment, the blanket

jacket, if present, surrounds the second neutron reflector jacket. In this embodiment, the second fuel salt jacket is formed by a plurality of tubular fuel salt vessels, each tubular fuel salt vessel extending axially from a fuel salt inlet to a fuel salt outlet for passage of a molten fuel salt 21, the plurality of tubular vessels being assembled to form a second cylindrical fuel salt jacket surrounding the moderator and neutron reflector jacket. The second neutron reflector jacket is formed by a plurality of tubular moderator and neutron reflector vessels. Each tubular moderator neutron reflector vessel extending axially from a liquid moderator and neutron reflector inlet to a liquid moderator and neutron reflector outlet for passage of a liquid moderator and neutron reflector.

In an embodiment (not shown in the Figs.) the nuclear reactor core 1 is provided with movable neutron absorbing control elements between vessels of the jackets, preferably between vessels making up the fuel salt jacket and/or between vessels forming a jacket that surrounds the fuel jacket or is surrounded by the fuel jacket. The neutron absorbing control elements are similar to control rods, except that the neutron absorbing control elements are not rod shaped but are shaped like a plate or wedge to match the adjacent vessels in the jacket concerned, and are arranged such that they can be inserted into and retracted from the nuclear reactor core 1, for controlling nuclear reactivity of the nuclear reactor core.

In an embodiment, the reactivity of the nuclear chain reaction of the nuclear reactor core 1 is controlled by controlling

the fuel salt temperature in the fuel salt jacket to remain at a temperature between approximately 600 and 700°C, and controlling the moderator and neutron reflector liquid in the central moderator and neutron reflector vessel 10 to remain at a temperature between approximately 20 and 90°C. Preferably also the temperature of the moderator and neutron reflector liquid 21 in the moderator and neutron reflector jacket 20 is controlled to remain at a temperature between approximately 20 and 90°C.

10

The method of controlling the reactivity of the nuclear chain reaction and temperature household of the nuclear reactor core 1 may further comprise:

- adjusting the liquid level of the liquid moderator and neutron reflector 11 in the central moderator and neutron reflector vessel 10, and/or
- adjusting the liquid level of the liquid moderator and neutron reflector 31 in the moderator and neutron reflector jacket 30, and/or
- adjusting the liquid level of the molten blanket salt 41 in the blanket salt jacket 40, and/or
- adjusting temperature of fuel salt 21 through the fuel jacket 20, and/or
- adjusting temperature of the liquid moderator and neutron reflector 11 in the center moderator and neutron reflector vessel 10, and/or
- adjusting temperature of the liquid moderator and neutron reflector 31 in the center moderator and neutron reflector jacket 30, and/or
- adjusting the chemical composition of the fuel salt 21, and/or

- adjusting the chemical composition of the liquid moderator and neutron reflector 11, 31, and/or
- adjusting the position of a control rod that is at least partially inserted in a lumen in the center moderator and neutron reflector vessel 10, and/or
- adjusting the position of a control rod that is at least partially inserted between fuel salt vessels 20, and/or
- adjusting the position of a control rod that is at least partially inserted between moderator and neutron absorption vessels 30.

Increasing the liquid level of the liquid moderator and neutron reflector in the central moderator and neutron reflector vessel 10, increases the reactivity of the nuclear chain reaction by providing more moderation and or reflection, and vice versa.

Increasing the liquid level of the liquid moderator and neutron reflector in the moderator and neutron reflector jacket 30, increases the reactivity of the nuclear chain reaction by providing more moderation and or reflection and or less neutron leakage, and vice versa.

Increasing the liquid level of the molten blanket salt 41 in the blanket salt jacket 40, if present, increases the reactivity of the nuclear chain reaction by providing more reflection and or less neutron leakage, and vice versa.

Increasing the temperature of the fuel salt in the fuel salt jacket 20, decreases the reactivity of the nuclear chain reaction by providing a lower average density in the nuclear reactor core and thus presence of less fissile fuel, and vice versa.

Increasing the temperature of the liquid moderator and neutron reflector in the center moderator and neutron reflector vessel 10, decreases the reactivity of the nuclear chain reaction by providing a lower average density in the nuclear reactor core and thus more moderation and or reflection, and vice versa.

Increasing the temperature of the liquid moderator and neutron reflector in the moderator and neutron reflector jacket, decreases the reactivity of the nuclear chain reaction by providing a lower average density in the nuclear reactor core and thus more moderation and or reflection and or more neutron leakage, and vice versa.

Adding fissile material to the fuel salt 21 increases the reactivity of the nuclear chain reaction, and vice versa.

Adding fertile material, e.g. thorium, or neutron absorbing material to the fuel salt 21 decreases the reactivity of the nuclear chain reaction, and vice versa.

Adding fertile material, e.g. thorium, or neutron absorbing material to the moderator and reflector liquid 11, 31 decreases the reactivity of the nuclear chain reaction, and vice versa.

Temperatures of the respective moderator liquid 11,31 or fuel salt 21 are changed by e.g.:

- increasing the flow rate resulting a smaller temperature drop across the nuclear reactor core 1 and thus a higher or lower density, and/or
- adjusting the cooling rate of the moderator liquids 11,31 or fuel salt 21 outside the nuclear reactor core 1 resulting in shifting the average temperature over the nuclear reactor core 1 and thus higher or lower densities.

Liquid levels of the respective moderator liquid 11,31 or fuel salt 21 or molten blanket salt 41 is changed by e.g. adjusting the rate at which liquid is pumped in the respective vessel or jacket from the top and the rate at which the liquid is passively allowed to drained from the bottom.

In an embodiment (not shown), a controller, e.g. an electronic control unit is coupled to the elements of the molten salt nuclear reactor 100 and configured to control the reactivity of the nuclear reactor core 1 in accordance with the measures described above.

The various aspects and implementations have been described in conjunction with various embodiments herein. However, other variations to the disclosed embodiments can be understood and effected by those skilled in the art in practicing the claimed subject-matter, from a study of the drawings, the disclosure, and the appended claims. In the claims, the word "comprising" does not exclude other elements or steps, and the indefinite article "a" or "an" does not exclude a plurality. The mere fact that certain measures are recited in mutually different dependent claims does not indicate that a combination of these measures cannot be used to advantage.

The reference signs used in the claims shall not be construed as limiting the scope. Unless otherwise indicated, the drawings are intended to be read (e.g., cross-hatching, arrangement of parts, proportion, degree, etc.) together with the specification, and are to be considered a portion of the

entire written description of this disclosure. As used in the description, the terms "horizontal", "vertical", "left", "right", "up" and "down", as well as adjectival and adverbial derivatives thereof (e.g., "horizontally", "rightwardly", "upwardly", etc.), simply refer to the orientation of the illustrated structure as the particular drawing figure faces the reader. Similarly, the terms "inwardly" and "outwardly" generally refer to the orientation of a surface relative to its axis of elongation, or axis of rotation, as appropriate.

PATENTKRAV:

1. Atomreaktorkerne (1) til en smeltetsaltatomreaktor (100), hvilken atomreaktorkerne (1) omfatter:

- 5 - en rørformet, cylindrisk centermoderator- og neutronreflektorbeholder (10) til passage af en væskemoderator og neutronreflektor (11), der strækker sig aksialt fra et væskemoderator- og neutronreflektorindløb (12) af centermoderator- og neutronreflektorbeholderen (10) til et
- 10 væskemoderator- og neutronreflektorudløb (13) af centermoderator- og neutronreflektorbeholderen (10),
- en flerhed af rørformede brændselssaltbeholdere (20), hvor hver rørformet brændselssaltbeholder (20) strækker sig aksialt fra et brændselssaltindløb (22) af den rørformede
- 15 brændselssaltbeholder (20) til et brændselssaltudløb (23) af den rørformede brændselssaltbeholder (20) til passage af et smeltet brændselssalt (21),
- en flerhed af rørformede moderator- og neutronreflektorbeholdere (30), hvor hver rørformet moderator-
- 20 og neutronreflektorbeholder (30) strækker sig aksialt fra et væskemoderator- og neutronreflektorindløb (32) af moderator- og neutronreflektorbeholderen (30) til et væskemoderator- og neutronreflektorudløb (33) af moderator- og neutronreflektorbeholderen (30) til passage af en væskemoderator
- 25 og neutronreflektor (31),
- kendetegnet ved, at
- flerheden af rørformede brændselssaltbeholdere (20) er samlet for at danne en cylindrisk brændselssaltkappe, der omgiver centermoderator- og neutronreflektorbeholderen (10),
- 30 flerheden af rørformede moderator- og neutronreflektorbeholdere (30) er samlet for at danne en cylindrisk moderator- og neutronreflektorkappe, der omgiver den cylindriske brændselssaltkappe, og

ved, at den cylindriske centermoderator- og neutronreflektorbeholder (10) har et størst indvendigt tværsnitsareal medialt mellem væskemoderator- og neutronreflektorindløbet (12) af centermoderator- og neutronreflektorbeholderen (10) og væskemoderator- og neutronreflektorudløbet (13) af moderator- og neutronreflektorbeholderen (10).

2. Atomreaktorkerne (1) ifølge krav 1, hvor det indvendige tværsnitsareal af centermoderator- og neutronreflektorbeholderen tiltager gradvist mod den mediale position mellem væskemoderator- og neutronreflektorindløbet (12) og væskemoderator- og neutronreflektorudløbet (13), idet det indvendige tværsnitsareal fortrinsvis varierer kontinuerligt mellem moderator- og neutronreflektorindløbet (12) og moderator- og neutronreflektorudløbet (13).

3. Atomreaktorkerne (1) ifølge krav 1 eller 2, hvor den cylindriske centermoderator- og neutronreflektorbeholder (10) og/eller væskemoderator- og neutronreflektorbeholderne (30) har en væg af en metallegering, fortrinsvis jernlegering, mest fortrinsvis rustfrit stål, zirconiumlegering eller keramisk kompositmateriale, og hvor brændselssaltbeholderne (20) fortrinsvis har en væg af keramisk kompositmateriale, en metallegering, en jernlegering, fortrinsvis rustfrit stål, eller nikkellegering, fortrinsvis Inconel eller Hastelloy.

4. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor hver brændselssaltbeholder (20) i brændselssaltkappen i alt væsentligt er identisk med de andre brændselssaltbeholdere (20) i brændselssaltkappen.

5. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor hver moderator- og

neutronreflektorbeholder (30) i moderator- og neutronreflektorkappen i alt væsentligt er identisk med de andre moderator- og neutronreflektorbeholdere (30) i moderator- og neutronreflektorkappen.

5

6. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor en første cylindrisk isoleringskappe er tilvejebragt mellem centermoderator- og neutronreflektorbeholderen (10) og brændselssaltkappen, hvilken første isoleringskappe fortrinsvis er dannet af ét eller flere første isoleringselementer (15), hvilke første isoleringselementer (15) fortrinsvis er formet som sektorer af den første cylindriske isoleringskappe eller formet som et bånd, der er spiralviklet omkring centermoderator- og neutronreflektorbeholderen (10), og hvilke første isoleringselementer (15) fortrinsvis er fremstillet af et grafitfilt eller aerogelimprægneret isoleringsgrafitfilt.

7. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor en anden cylindrisk isoleringskappe er tilvejebragt mellem brændselssaltkappen og moderator- og neutronreflektorkappen, hvilken anden isoleringskappe fortrinsvis er dannet af ét eller flere anden isoleringselementer (25), hvilke anden isoleringselementer (25) fortrinsvis er formet som sektorer af den anden cylindriske isoleringskappe eller formet som et bånd, der er spiralviklet omkring brændselssaltkappen, og hvilke anden isoleringselementer (25) fortrinsvis er fremstillet af et grafitfilt eller aerogelimprægneret isoleringsgrafitfilt.

30

8. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, der omfatter en flerhed af rørformede dæksaltbeholdere (40), hvor hver rørformet dæksaltbeholder (40) fortrinsvis strækker sig aksialt fra et indløb af

dæksaltbeholderen (40) til et udløb fra dæksaltbeholderen (40) til passage af et smeltet dæksalt (41), hvor flerheden af rørformede dæksaltbeholdere (40) er samlet for at danne en cylindrisk dæksaltkappe, der omgiver moderator- og
5 neutronreflektorkappen.

9. Atomreaktorkerne (1) ifølge krav 8, hvor en tredje cylindrisk isoleringskappe er tilvejebragt mellem neutronmoderator- og reflektorkappen og dæksaltkappen, hvilken tredje
10 isoleringskappe fortrinsvis er dannet af ét eller flere tredje isoleringselementer (35), hvilke tredje isoleringselementer (35) fortrinsvis er formet som sektorer af den tredje cylindriske isoleringskappe eller formet som et bånd, der er spiralviklet omkring moderator- og neutronreflektorkappen, og
15 hvilke tredje isoleringselementer (35) fortrinsvis er fremstillet af et grafitfilt eller aerogelimprægneret isoleringsgrafitfilt.

10. Atomreaktorkerne (1) ifølge et hvilket som helst af de
20 foregående krav, hvor centermoderator- og neutronreflektorbeholderen (10), brændselssaltbeholderne (20), moderator- og neutronreflektorbeholderne (30) og/eller dæksaltbeholderne (40) hver er samlet af to dele, der er adskilt af et symmetriplan (M), der strækker sig radiale fra den mediale
25 position, hvor brændselsdækbholderne (40) fortrinsvis har en væg af keramisk kompositmateriale, en metallegering, en jernlegering, fortrinsvis rustfrit stål, eller nikkellegering, fortrinsvis Inconel eller Hastelloy.

30 11. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor centermoderator- og neutronreflektorbeholderen (10) har en variabel indvendig og udvendig radius R_{11} og R_{12} , hvor den indvendige og udvendige radius R_{11} og R_{12} fortrinsvis er størst ved eller i nærheden af

den mediale position og gradvist aftager fra den mediale position.

12. Atomreaktorkerne (1) ifølge et hvilket som helst af de
5 foregående krav, hvor brændselssaltkappen er formet som en hul
cylinder med en variabel indvendig og udvendig radius R_{21} og
 R_{22} , og hvor brændselssaltbeholderne (20) fortrinsvis er formet
som en cirkelformet, cylindrisk sektor med en variabel
henholdsvis indvendig og udvendig radius R_{21} og R_{22} , og en given
10 sektorvinkel A_2 , hvor den indvendige og udvendige radius R_{21} og
 R_{22} fortrinsvis er størst ved eller i nærheden af den mediale
position og gradvist aftager fra den mediale position.

13. Atomreaktorkerne (1) ifølge et hvilket som helst af de
15 foregående krav, hvor moderator- og neutronreflektorkappen er
formet som en hul, cirkelformet cylinder med en variabel
indvendig og udvendig radius R_{31} og R_{32} , og hvor
moderatorneutronreflektorbholderen (30) fortrinsvis er formet
som en cirkelformet, cylindrisk sektor med en variabel
20 henholdsvis indvendig og udvendig radius R_{31} og R_{32} , og en given
sektorvinkel A_3 , hvor den indvendige og udvendige radius R_{31} og
 R_{32} fortrinsvis er størst ved eller i nærheden af den mediale
position og gradvist aftager fra den mediale position.

25 14. Atomreaktorkerne (1) ifølge et hvilket som helst af kravene
8 til 13, hvor dæksaltkappen er formet som en hul, cirkelformet
cylinder med en variabel indvendig og udvendig radius R_{41} og
 R_{42} , og hvor dæksaltbeholderne (40) fortrinsvis er formet som
en cirkelformet, cylindrisk sektor med en variabel henholdsvis
30 indvendig og udvendig radius R_{41} og R_{42} , og en given sektorvinkel
 A_4 , hvilken indvendig og udvendig radius R_{41} og R_{42} fortrinsvis
er størst ved eller i nærheden af den mediale position og
gradvist aftager fra den mediale position.

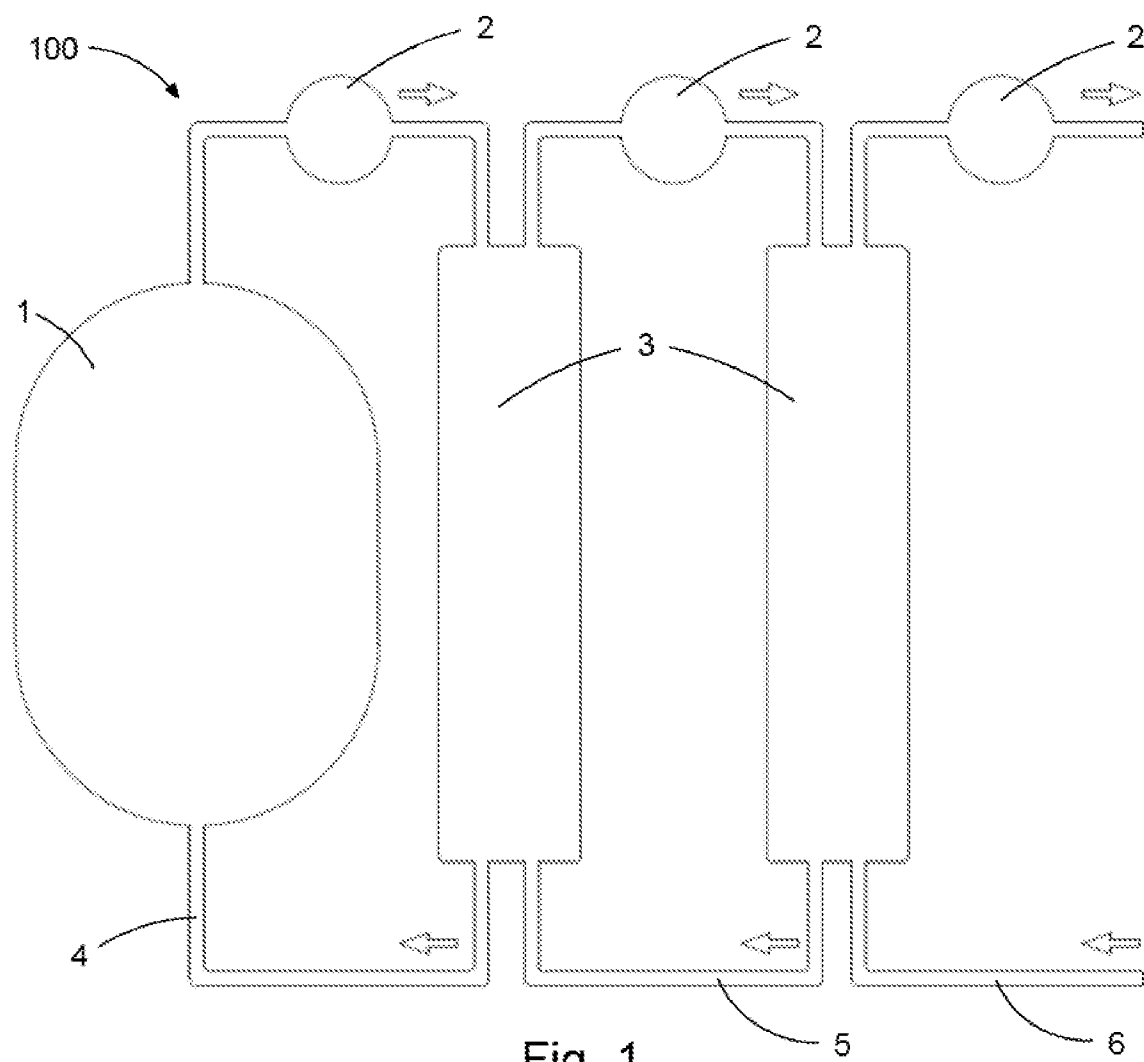
15. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor brændselssaltkappen, neutronmoderator- og reflektorkappen og/eller dækkappen i alt væsentligt er periferisk kontinuerlige kapper, fortrinsvis dannet af tilsvarende sider af de rørformede beholdere (20, 30, 40), der danner disse kapper, som rører hinanden langs en væsentlig del af deres aksiale udstrækning.

16. Atomreaktorkerne (1) ifølge et hvilket som helst af de foregående krav, hvor centermoderator- og neutronreflektorbeholderen (10) er segmenteret og dannet af en flerhed af rørformede moderatorbeholdere, hvor hver moderatorbeholder strækker sig aksialt fra et moderatorindløb til et moderatorudløb til passage af en moderator, hvor flerheden af rørformede moderatorbeholdere er samlet for at danne centermoderatorbeholderen (10).

17. Atomreaktorkerne (1) ifølge krav 16, hvor centermoderator- og neutronreflektorbeholderen (10) har en periferisk udvendig væg og en periferisk indvendig væg, hvor den periferiske indvendige væg fortrinsvis frembringer et lumen til modtagelse af en styrestang.

18. Smeltetsaltatomreaktor (100), der omfatter en atomreaktorkerne (1) ifølge et hvilket som helst af kravene 1 til 17.

1/6



2/6

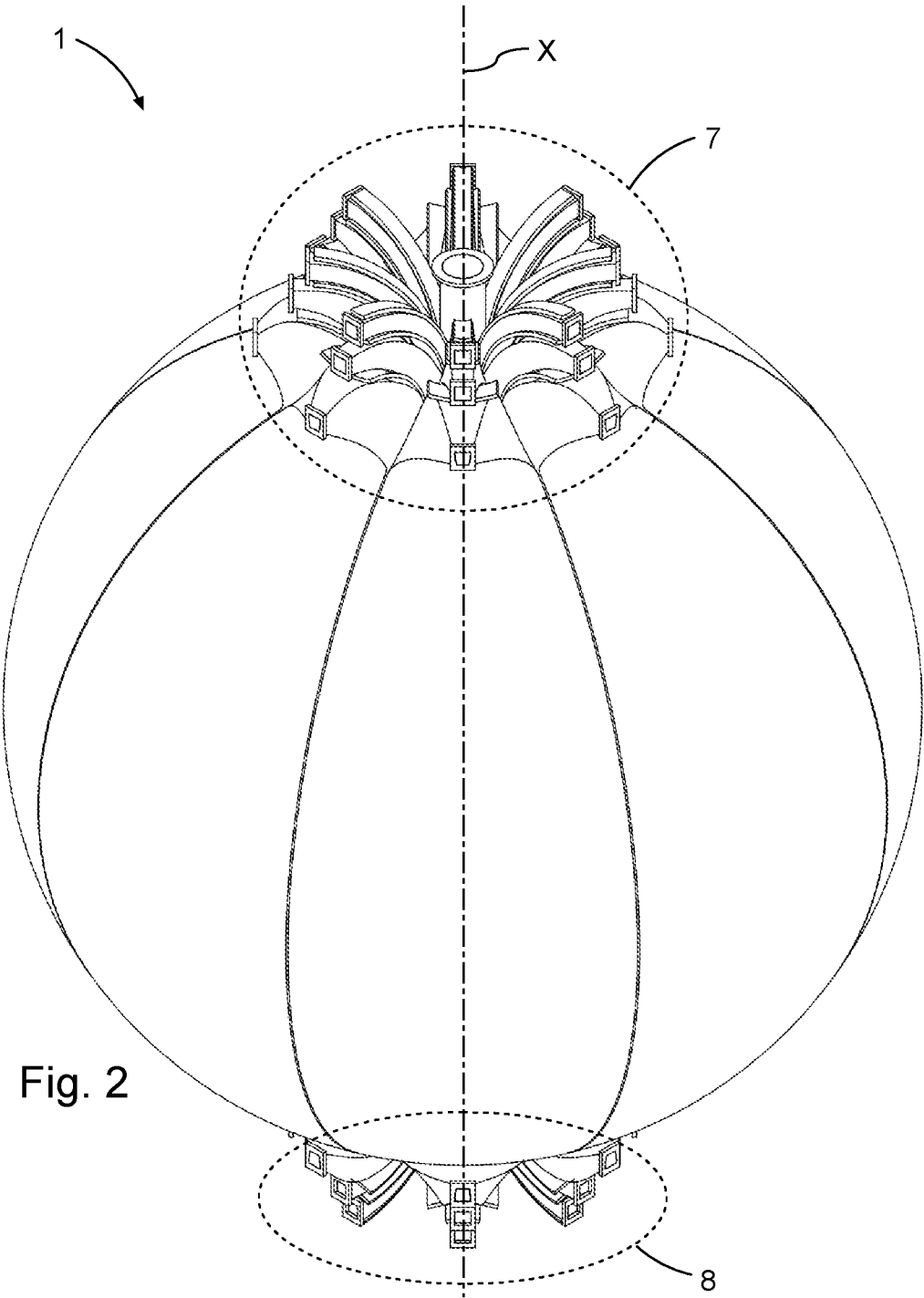
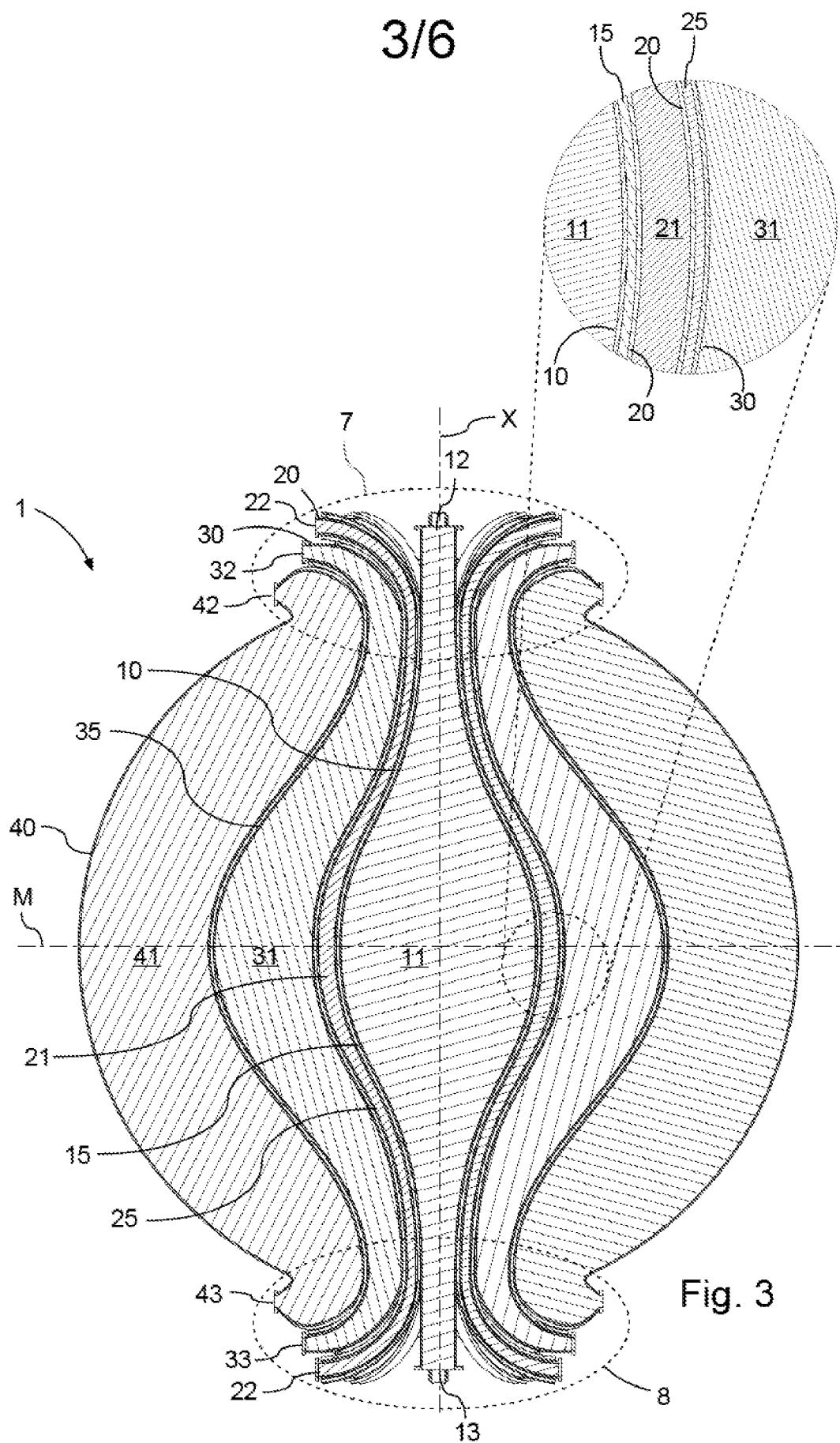
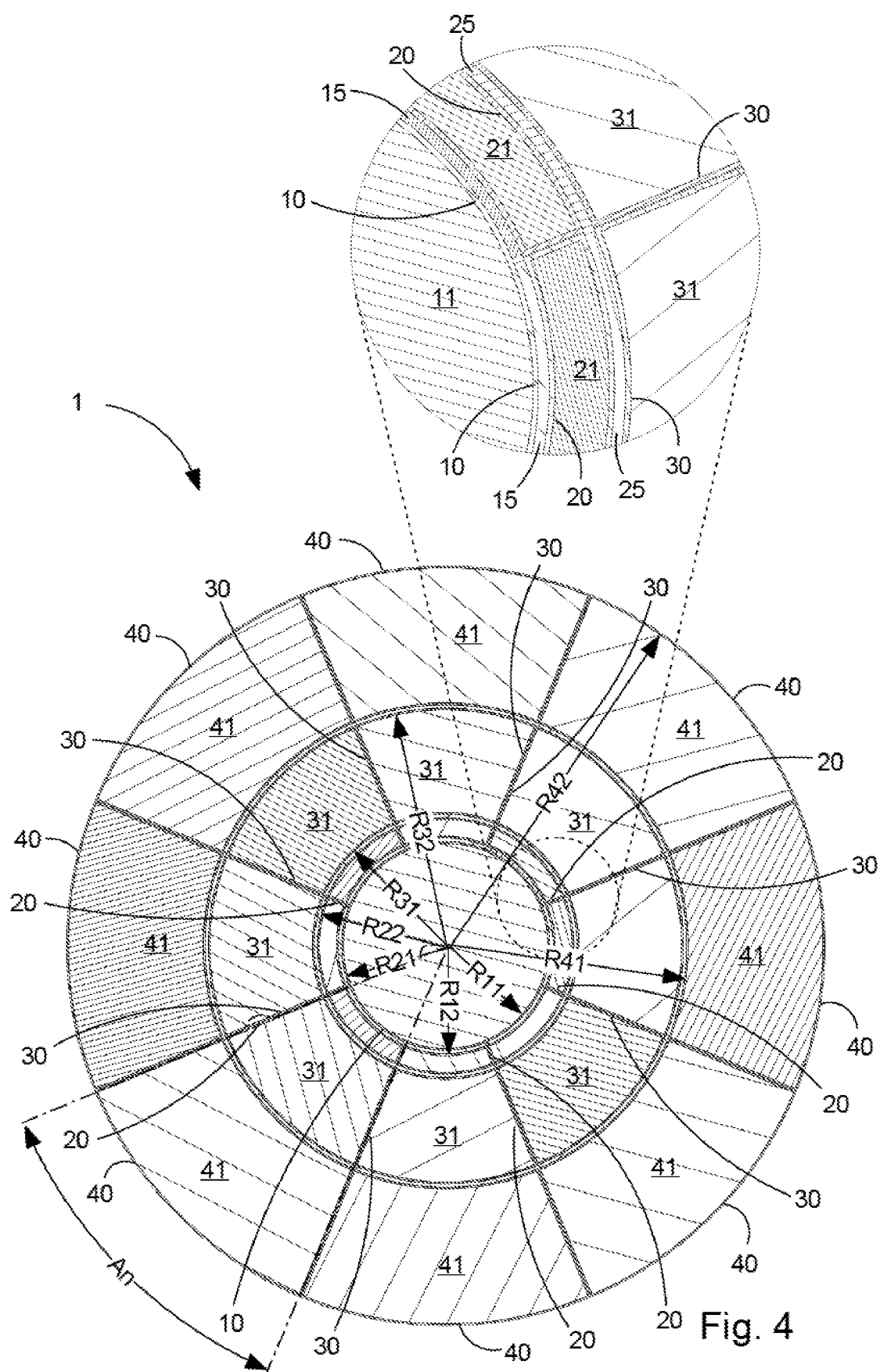


Fig. 2

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5/6

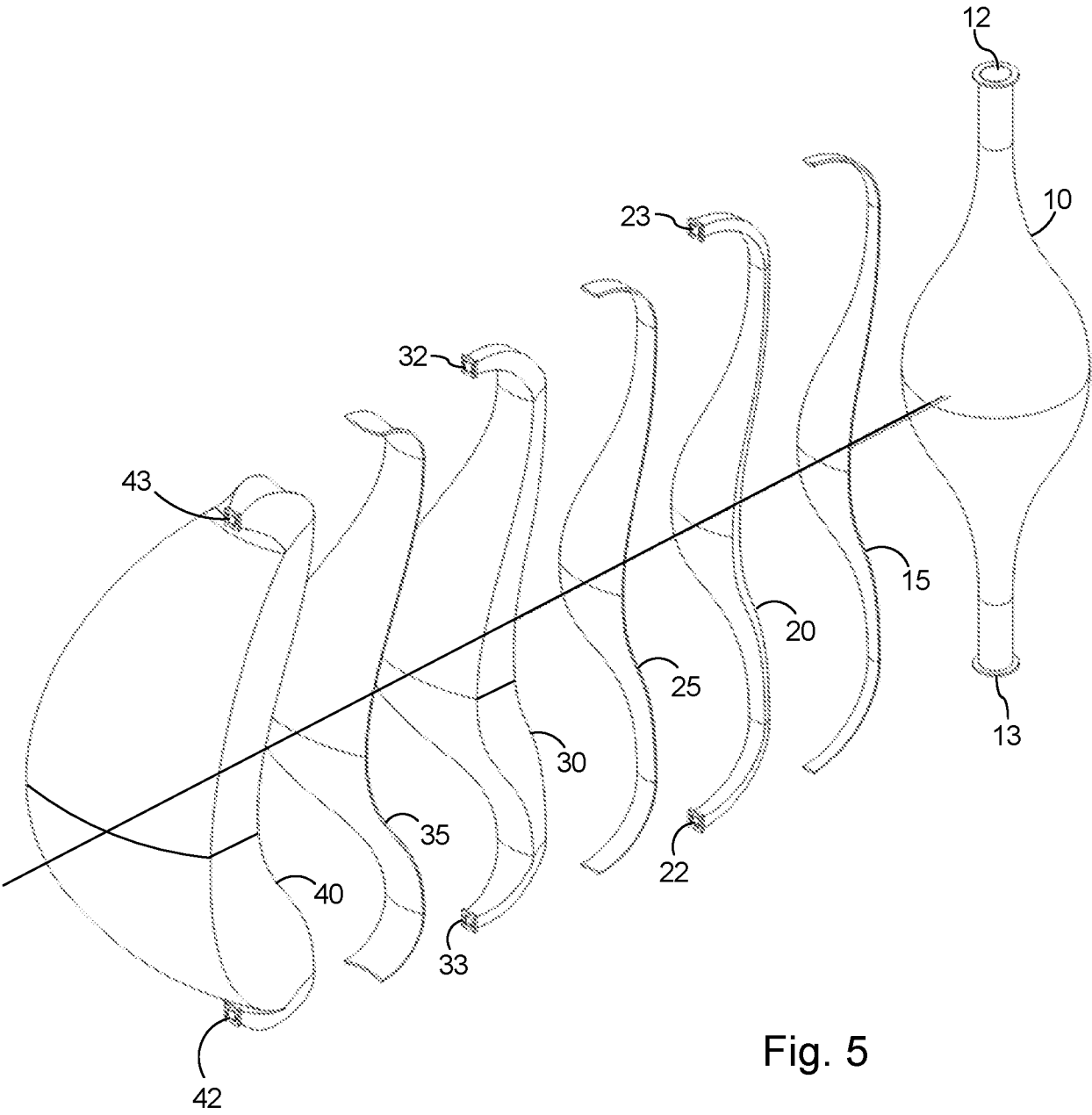


Fig. 5

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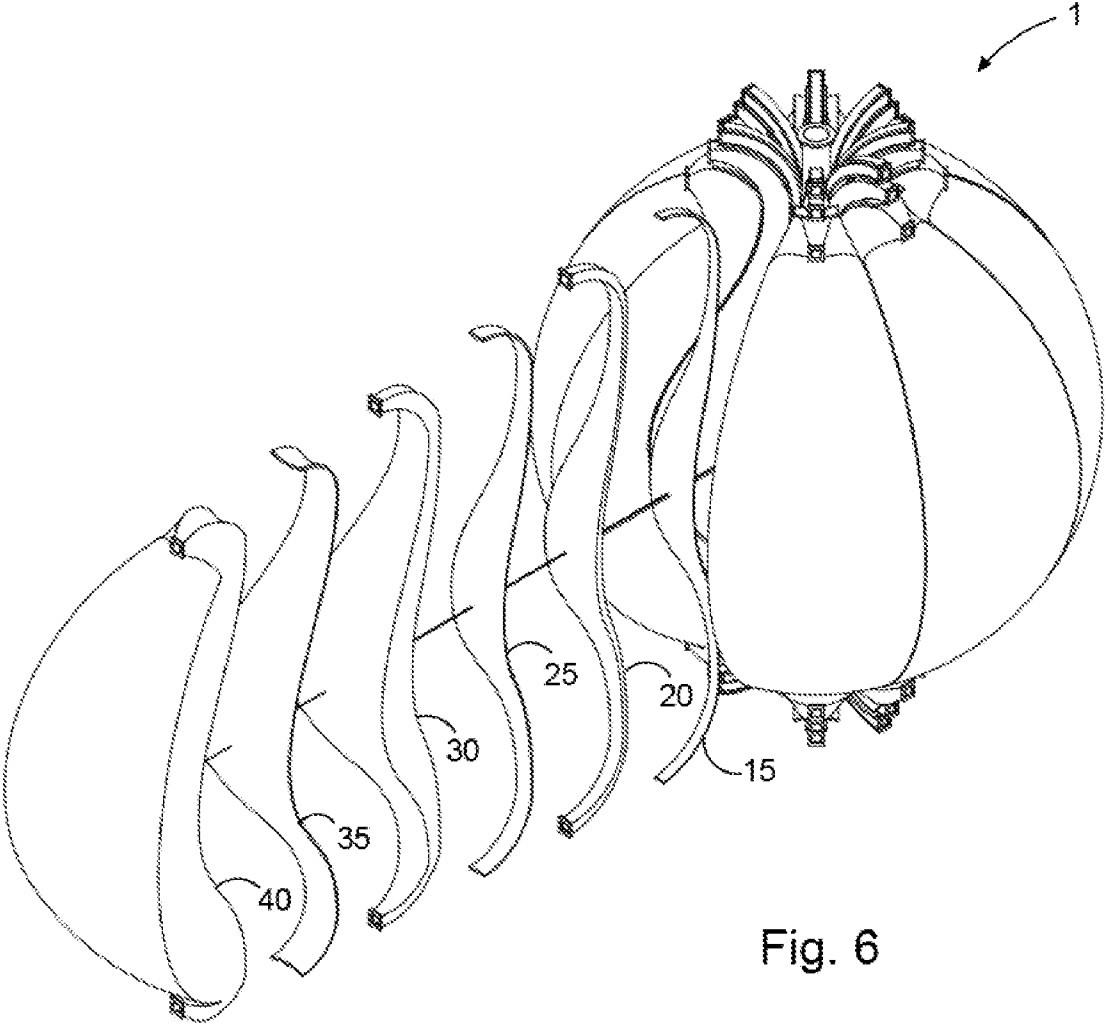


Fig. 6

SEARCH REPORT - PATENT		Application No. PA 2021 70280
1. ☐ Certain claims were found unsearchable (See Box No. I). 2. ☐ Unity of invention is lacking prior to search (See Box No. II).		
A. CLASSIFICATION OF SUBJECT MATTER G21C 5/14 (2006.01), G21C 1/22 (2006.01), G21C 13/04 (2006.01) According to International Patent Classification (IPC)		
B. FIELDS SEARCHED PCT-minimum documentation searched (classification system followed by classification symbols) IPC&CPC: G21C Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched DK, NO, SE, FI: IPC-classes as above. Electronic database consulted during the search (name of database and, where practicable, search terms used) EPODOC, WPI, FULL TEXT: English		
C. DOCUMENTS CONSIDERED TO BE RELEVANT		
Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant for claim No.
X A	<u>CN 112259263</u> A (SHANGHAI INST APPLIED PHYSICS CAS) 22 nd January 2022, see in particular abstract; paras. 0012-0013, 0072-0073, 0078-0080; and Figures 1-2, 5 (The above cited paragraph numbers are according to the numeration used in the enclosed machine translation, rather than that in the original document.)	1, 3, 6-10, 15-18 2, 4-5, 11-14
A	<u>US 2018/0068750</u> A1 (CISNEROS et al.) 8 th March 2018, see abstract, Figures 1-11 and paragraphs 0072-0089	1-18
A	<u>WO 2018/071081</u> A2 (ELYSIUM INDUSTRIES LTD) 19 th April 2018, see abstract, Figures 1-4 and paragraphs 0079-0104	1-18
A	<u>US 2017/0301416</u> A1 (HANSEN) 19 th October 2017, see abstract, Figures 4A-4O and paragraphs 0105-0121.	1-18
☐ Further documents are listed in the continuation of Box C.		
* Special categories of cited documents: "A" Document defining the general state of the art which is not considered to be of particular relevance. "D" Document cited in the application. "E" Earlier application or patent but published on or after the filing date. "L" Document which may throw doubt on priority claim(s) or which is cited to establish the publication date of another citation or other special reason (as specified). "O" Document referring to an oral disclosure, use, exhibition or other means.	"P" Document published prior to the filing date but later than the priority date claimed. "T" Document not in conflict with the application but cited to understand the principle or theory underlying the invention. "X" Document of particular relevance; the claimed invention cannot be considered novel or cannot be considered to involve an inventive step when the document is taken alone. "Y" Document of particular relevance; the claimed invention cannot be considered to involve an inventive step when the document is combined with one or more other such documents, such combination being obvious to a person skilled in the art. "&" Document member of the same patent family.	
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SEARCH REPORT – PATENT		Application No. PA 2021 70280
C (Continuation). DOCUMENTS CONSIDERED TO BE RELEVANT		
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SUPPLEMENTAL BOX

Continuation of Box [.]